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24K-0705 | BCS-4J.

Data Base System | Assignment #03.

Question 1)

- 1) π Platform_name (σ Difficulty-level = "Advanced" $\wedge$ Enroll.Date Like "%2026%") (Platform $\bowtie$ Enrollment $\bowtie$ Courses))
 - It joins Platform, Enrollment and course tables to filter by Year 2026 & "Advanced" difficulty, then projects platform names.
- 3) π Platform_name (Platform) - π Platform_name (Platform $\bowtie$ Instructor)
 - It subtracts platform's that have a match in the instructor table from the set of all platforms.
- 2) π Instructor_Name (σ Category = "Data Science" $\wedge$ Status = "Complete") (Instructors $\bowtie$ Course-Activity $\bowtie$ Enrollment $\bowtie$ Course))
 - It links instructors to their activity and course category to find those who finished "DS" courses.
- 4) π Course_Name (σ count > 1) (π Course_Id count (Platform_Id) $\rightarrow$ count (ENROLLMENT)) $\bowtie$ Course)
 - Groups enrollments by course-id, counts platforms, filter by counts > 1, and joins with course for names.
- 5) π . Course_name, Platform_name, completion_rate (σ Category = "AI" $\wedge$ Completion_Rate > 85 (Courses $\bowtie$ Enrollment $\bowtie$ Platform))
 - It joins 3 tables to retrieve AI courses with a rate above 85% along with platform names.

Question 2

- 1) Select title
from Book
where isAvailable = 0 And Price > 1000;
(Title of books that are unavailable & cost over 1000).
- 2) Select title
from Book
where Genre = "Fiction" AND Price > 800 OR Genre = "History";
- 3) Select title, price
from Book
where Genre = "Science" And isAvailable = 1;
- 4) Select title, price
from Book
where Genre = "Science" And isAvailable = 1;
- 5) Select Publisher-ID
from Publishers
Join Book using (bookID)
where title = "AI Basics";
- 6) Select s.name, Ext-LibName
from staffs
Join Library L
on S.libID = L.libID.
where s.address = L.city;
- 7) Select staff-id from staff.
where salary < 6000;
- 8) Select Publish-id.
from Publisher
where city = "Karachi"
Union.
Select PublisherID
from Publisher
where city = "Isb"
- 9) Select title
from Book
Except
Select title from Book
Join Publishers
using bookID.

Question 3:

- a) $R_1(A), R_2(A), A := A - N, W_1(A)$
 $A := A + M, W_2(A)$
- b) 1) Step 1: $R_1(A)$ is followed by $W_2(A)$ (conflict = $T_1 \rightarrow T_2$)
2) Step 2: $R_2(A)$ is followed by $W_1(A)$ (conflict = $T_2 \rightarrow T_1$).
- c) No, there is a cycle ($T_1 \rightarrow T_2$ & $T_2 \rightarrow T_1$).
- d) $R_1(A), W_1(A), R_2(A), W_2(A)$
- e) This causes the lost update problem. One transaction's change are overwritten by the other, leading to incorrect bank balance.

Question 4:

- 1) $S_1: T_1 \rightarrow T_3$ (on x), $T_2 \rightarrow T_4$ (on y), $T_3 \rightarrow T_1$ (on x),
 $T_4 \rightarrow T_2$ (on y)
 $S_2: T_5 \rightarrow T_1$ (on x), $T_1 \rightarrow T_2$ (on y), $T_2 \rightarrow T_4$ (on y)
- 2) S_1 :- Not serializable
 S_2 :- Serializable.
- 3) $T_3 \rightarrow T_1 \rightarrow T_2 \rightarrow T_4$.

Question 5:

- 1) T_2 overwrites the changes made by T_1 because both read the initial balance before either of them had written their results.
- 2) 7000
- 3) $5000 - 1000 + 2000 = 6000$.

Question 6.

a) Concept:- Every transaction gets a unique "Ticket number" upon arrival. Operations must execute in ticket order, if an older transaction tries to access data already modified by a newer one, it is rejected.

• Problems :-

Starvation. A long transaction might be repeatedly reestrated because newer, faster transaction keep "violating" the order.

• Strict Timestamp ordering :-

Ensures that if a transaction writes to an item no other transaction can read or write that item until the first one commits or abort. This prevents "messy" cascading rollbacks.

b) Situation :-

T_1 locks account A & wants B. Simultaneously T_2 locks account B & wants A. Both waits for the other to release the lock.

Resolution :-

The system maintains a "wait-for" graph. If a cycle is detected; the system aborts one of the transactions to break the cycle & allow the other to finish.